

QuadrupolarDFT

User Manual

Ab initio electric-field gradients and finite-temperature NQR parameters

QuadrupolarDFT is the electric-field-gradient (EFG) workspace within the MR-SpinDynamics repository. It converts first-principles EFG tensors into nuclear quadrupolar coupling constants, asymmetry parameters, and zero-field NQR transition frequencies, and corrects those 0 K static quantities to a finite temperature through harmonic vibrational averaging driven by a finite-displacement ABINIT DFPT workflow.

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1 Scope and Physical Background

QuadrupolarDFT addresses one link in the magnetic-resonance modelling chain: it turns an *ab initio* electronic-structure result into the parameters that control a nuclear quadrupole resonance (NQR) experiment.

A nucleus with spin $I \geq 1$ carries an electric quadrupole moment Q that couples to the electric-field-gradient (EFG) tensor $V_{ij} = \partial^2 V / \partial x_i \partial x_j$ produced by the surrounding charge distribution. The EFG is symmetric and, away from the nucleus, traceless (Laplace's equation), so in its principal-axis system it is described by two numbers: the largest-magnitude component V_{zz} and the asymmetry parameter

$$\eta = \frac{V_{xx} - V_{yy}}{V_{zz}}, \quad |V_{zz}| \geq |V_{yy}| \geq |V_{xx}|, \quad 0 \leq \eta \leq 1. \quad (1.1)$$

The quadrupolar coupling constant follows from V_{zz} and the nuclear quadrupole moment,

$$C_Q = \frac{eQV_{zz}}{h}, \quad (1.2)$$

and the zero-field NQR transition frequencies follow from C_Q , η and the spin I by diagonalizing the quadrupole Hamiltonian

$$\frac{H_Q}{h} = \frac{C_Q}{4I(2I-1)} [3I_z^2 - I(I+1) + \eta(I_x^2 - I_y^2)]. \quad (1.3)$$

For spin $I = 1$ (e.g. ^{14}N) this gives the familiar triplet

$$\nu_{\pm} = \frac{3}{4}C_Q \left(1 \pm \frac{\eta}{3}\right), \quad \nu_0 = \nu_+ - \nu_- = \frac{1}{2}C_Q \eta. \quad (1.4)$$

1.1 Why a finite-temperature correction is needed

A static DFT calculation evaluates V_{ij} at a single, fixed nuclear configuration at 0 K. Measured NQR frequencies, however, are taken at finite temperature and are strongly temperature dependent: in molecular crystals the lines typically *fall* as temperature rises (the Bayer law). Two physically distinct effects separate the static DFT number from the measurement:

1. **Thermal expansion / geometry.** The lattice constants and internal coordinates at temperature T differ from the relaxed 0 K cell. This is a static effect: it is captured by evaluating the EFG on a temperature-appropriate structure.
2. **Vibrational averaging.** The nucleus samples a thermal distribution of positions through the phonons. The measured EFG is the tensor averaged over that motion, including zero-point motion, which is present even at 0 K. For molecular crystals this dynamic term usually dominates: low-frequency librations reorient the EFG principal axes and reduce the line frequencies.

QuadrupolarDFT provides the static layer (Chapter 3) and a harmonic treatment of the vibrational term (Chapters 4–5).

2 Installation and Conventions

2.1 Installation

The package depends only on NumPy. From the `QuadrupolarDFT/` directory:

```
python -m pip install -e .
python -m pip install -e ".[dev]"      # adds ruff for linting
python -m unittest discover -s tests
```

On a system whose Python is externally managed (PEP 668), the package can also be used without installation by setting `PYTHONPATH` to the `src/` directory, since NumPy is the only requirement:

```
export PYTHONPATH="$PWD/src"
python3 examples/abinit/efg_temperature.py --help
```

2.2 Tensor and unit conventions

`EFGTensor` stores the symmetric tensor in SI units (V/m^2) and sorts the principal components as $|V_{zz}| \geq |V_{yy}| \geq |V_{xx}|$. It accepts input in ABINIT atomic units (the default), in SI, or in the $10^{21} \text{ V}/\text{m}^2$ scale, and exposes `vzz_si` and `eta`. Tensor averaging (`average_tensors`) operates on the tensor components and only then diagonalizes the result, so that asymmetry information survives the average.

2.3 Link to the spin-dynamics simulator

The companion `spin_dynamics` package parameterizes a quadrupolar site by its $\eta = 0$ transition frequency ν_Q . This is related to C_Q by the prefactor of Eq. (1.3):

$$\nu_Q = C_Q \cdot \frac{d}{4I(2I-1)} = \begin{cases} \frac{3}{4} C_Q & I = 1, \\ \frac{1}{2} C_Q & I = \frac{3}{2}, \end{cases} \quad (2.1)$$

with $d = 3$ for spin 1 and $d = 6$ for spin 3/2. The cross-project `mr_integration` layer uses this mapping to feed DFT parameters into the simulator and to compare predictions against the measured NQR database.

3 Static EFG Workflow

The static path produces C_Q , η and the NQR lines from an ABINIT EFG calculation.

1. `format_abinit_efg_block` emits the `nucsfq/quadmom` input lines that request an EFG calculation (PAW datasets are required; norm-conserving pseudopotentials are not suitable).
2. ABINIT is run on the input (see `examples/abinit/run_nano2_efg_wsl.sh`).
3. `parse_abinit_efg` parses the Electric Field Gradient Calculation blocks of the output into `AbinitEFGRecord` objects, one per atom, carrying C_Q , η , the principal values/axes, and the total EFG tensor.
4. `coupling_constant_hz` and `nqr_frequencies_hz` convert a tensor and a nuclear quadrupole moment into C_Q and the unique zero-field transition frequencies (Eqs. (1.2)–(1.3)).

`examples/parse_abinit_efg.py` demonstrates parsing an output file. The EFG components, not the total energy, must be converged against `ecut`, `pawecutdg` and the k -point mesh.

4 Finite-Temperature EFG

4.1 Harmonic vibrational averaging

Expanding the EFG tensor to second order in the mass-weighted phonon normal coordinates Q_k and averaging over the harmonic distribution gives

$$\langle V_{ij} \rangle(T) = V_{ij}^{\text{eq}} + \frac{1}{2} \sum_k \frac{\partial^2 V_{ij}}{\partial Q_k^2} \langle Q_k^2 \rangle(T), \quad (4.1)$$

where the mean-square amplitude of mode k (angular frequency ω_k) is

$$\langle Q_k^2 \rangle(T) = \frac{\hbar}{2\omega_k} \coth\left(\frac{\hbar\omega_k}{2k_B T}\right). \quad (4.2)$$

The $\coth$ factor equals $2n(\omega_k, T) + 1$ with n the Bose–Einstein occupation. It tends to 1 as $T \rightarrow 0$ (pure zero-point motion — which is why the static EFG is never exactly what is measured) and to $2k_B T / \hbar\omega_k$ in the classical high-temperature limit. These quantities live in `quadrupolar_dft.thermal`.

Tensor-frame averaging. The average in Eq. (4.1) is taken on the tensor components in a fixed crystal frame; the result is diagonalized only afterwards. Librational motion reorients the principal-axis system, so averaging V_{zz} or η directly would discard exactly that effect. As a consequence η is itself temperature dependent. `thermally_averaged_efg` performs the average and removes any residual trace (the EFG is traceless by Laplace’s equation; the finite-difference curvatures of Section 5.1 amplify rounding noise into a spurious trace, which is projected out). `efg_temperature_sweep` returns $C_Q(T)$, $\eta(T)$ and the lines at each requested temperature.

4.2 The analytic Bayer limit

Where per-mode EFG curvatures are not available, the temperature *form* can be validated directly against a measured line series with the single-libration Bayer–Kushida model

$$\nu(T) = \nu_0 \left[1 - a \coth\left(\frac{\hbar\omega}{2k_B T}\right) \right], \quad (4.3)$$

where a is a small dimensionless librational amplitude. `fit_bayer_single_mode` fits (ν_0, a, ω) to measured $\nu(T)$ data with a NumPy-only routine: for a fixed ω the model is linear in ν_0 and $\nu_0 a$, so a grid scan over ω with a linear least-squares solve at each point suffices.

5 The DFPT Finite-Displacement Workflow

The mode curvatures $\partial^2 V_{ij} / \partial Q_k^2$ in Eq. (4.1) come from displacing the structure along each phonon mode and recomputing the EFG. Because the DFT runs are expensive and run outside the package, the workflow separates into three stages with a calculation between each, preceded by an optional structure-relaxation stage (Stage 0) that puts the geometry at an energy minimum first. The CLI `examples/abinit/efg_temperature.py` drives all of them.

5.1 Normal-coordinate convention

A phonon eigenvector ε_k is mass-weighted and normalized ($\sum |\varepsilon_k|^2 = 1$). A normal-coordinate step δQ displaces atom i (mass m_i) by

$$\Delta \mathbf{r}_i = \delta Q \frac{\varepsilon_{k,i}}{\sqrt{m_i}}, \quad (5.1)$$

and the curvature is the central difference of the EFG tensors at $0, \pm \delta Q$,

$$\frac{\partial^2 V_{ij}}{\partial Q_k^2} \approx \frac{V_{ij}(+\delta Q) - 2V_{ij}(0) + V_{ij}(-\delta Q)}{\delta Q^2}. \quad (5.2)$$

These units are consistent with $\langle Q_k^2 \rangle$ from Eq. (4.2), so the product in Eq. (4.1) is an EFG. `normal_coordinate_step_si` chooses δQ from a target maximum Cartesian displacement (default 0.05 Å).

5.2 Stage 0 — structure relaxation (optional)

A static EFG is only physical at an energy minimum. On an unrelaxed geometry the residual forces appear as imaginary modes at Γ and bias the equilibrium η , so relaxing first is the single largest source of quantitative error in the chain. `efg_temperature.py relax` strips the EFG keywords from a converged static input and prepends a BFGS ionic-relaxation block (`ionmov 2, optcell 0, tolmxf`); `run_relax_wsl.sh` runs ABINIT. `efg_temperature.py relax-collect` then reads the relaxed geometry from the output footer (`-outvars: echo values of variables after computation, the same block abipy reads`) and writes a relaxed static EFG input.

```
python3 examples/abinit/efg_temperature.py relax \
--base examples/abinit/nano2_efg.abi --out runs/nano2_relax
bash examples/abinit/run_relax_wsl.sh runs/nano2_relax
python3 examples/abinit/efg_temperature.py relax-collect \
--base examples/abinit/nano2_efg.abi --abo runs/nano2_relax/relax.abo \
--out runs/nano2_relax
```

The product, `relaxed.abi`, is an ordinary static EFG input (EFG keywords intact) at the relaxed geometry; pass it as `-base` to Stages 1–2 in place of the hand-written input. The relaxation is internal-coordinates-only: the cell is held at the input value (`optcell 0`). For molecular crystals this is usually preferable to a full cell relaxation, which under GGA tends to over-expand the lattice and can worsen the EFG relative to the measured geometry; full cell relaxation (the quasi-harmonic thermal-expansion path) is a planned extension.

5.3 Stage 1 — phonons

`efg_temperature.py phonon` writes a two-dataset ABINIT input (ground state plus a Γ -point DFPT response) and the matching `anaddb` input, derived from a converged static EFG input. `run_phonon_wsl.sh` runs ABINIT, selects the DFPT derivative database (the largest `*_DDB`), and runs `anaddb` to produce the phonon frequencies and eigenvectors.

```
python3 examples/abinit/efg_temperature.py phonon \
    --base examples/abinit/nano2_efg.abi --out runs/nano2_ph
bash examples/abinit/run_phonon_wsl.sh runs/nano2_ph
```

The generated DFPT input is a starting template. Two settings are required for PAW + GGA phonons and are emitted automatically: `iscf2 7` (the response driver requires `iscf  $\leq 9$`) and `pawxcdev 0` (the GGA exchange correlation kernel is not implemented in the development scheme). Converge the tolerances and k -mesh for production use.

5.4 Stage 2 — displaced EFG runs

`efg_temperature.py displace` parses the `anaddb` eigenvectors (`parse_anaddb_modes`; acoustic and imaginary modes are dropped, and modes are paired with frequencies by mode number so that gaps in the eigenvector output do not misalign them), generates one displaced EFG input per $\pm$ mode plus a manifest, and writes positions as fractional xred coordinates. `run_finite_displacement_wsl.sh` runs ABINIT on every input.

```
python3 examples/abinit/efg_temperature.py displace \
    --base examples/abinit/nano2_efg.abi --anaddb runs/nano2_ph/anaddb.out \
    --target 2 --max-modes 6 --out runs/nano2_disp
bash examples/abinit/run_finite_displacement_wsl.sh runs/nano2_disp
```

`-target` is the 0-based index of the resonant nucleus; `-max-modes` restricts the calculation to the lowest-frequency librations, which dominate the temperature dependence. If the `anaddb` eigenvector layout differs from the parser's expectation, `-modes modes.json` supplies frequencies and eigenvectors directly.

5.5 Stage 3 — collect

`efg_temperature.py collect` parses the displaced EFG outputs, central differences the mode curvatures, runs the temperature sweep, and reports $C_Q(T)$, $\eta(T)$, the lines, and $d\nu/dT$.

```
python3 examples/abinit/efg_temperature.py collect \
    --workdir runs/nano2_disp --temperatures 0,77,150,300 --quadmom 0.02044
```

5.6 Running ABINIT in parallel

The phonon and EFG runs dominate the wall-clock cost, and ABINIT (MPI-built) parallelizes them efficiently over k -points. All the runner scripts source a small shared helper, `examples/abinit/abinit_parallel.sh`, and honor the `ABINIT_NP` environment variable:

- `ABINIT_NP` unset (or 1) — serial.
- `ABINIT_NP=N` — launch `mpirun -np N abinit`.
- `ABINIT_NP=auto` — choose N automatically (see below) and fall back to serial if no MPI runtime is found.

With the default k -point parallelism (`para1_kgb 0`), efficiency is good when the number of processes divides the number of k -points in the IBZ. The helper function `abinit_optimal_np` encodes this: it returns the largest divisor of n_{kpt} that is $\leq$ the core count. For the NaNO_2 cell ($n_{\text{kpt}} = 36$) on a 20-core machine that is $N = 18$ (two k -points per process), which scales near-ideally ($\sim 18\times$). `auto` detects n_{kpt} from an existing `.abo` or a quick `abinit -dry-run`, and the core count from `nproc`. Override the launcher with `ABINIT_MPIRUN` (e.g. `mpiexec`, or `mpirun -oversubscribe`). The driver `nano2_relaxation_study.sh` uses `auto` by default; pass `-np N` to pin a value or `-np 1` to force serial.

Parallelization changes only how ABINIT is launched, not the inputs, so it leaves all computed quantities (and any relaxed-vs-unrelaxed comparison) unchanged.

6 Worked Example: NaNO_2 ^{14}N

Sodium nitrite is the reference case for the workspace: its ^{14}N parameters appear in the DFT runs, the spin-dynamics simulator, and the measured NQR database. A six-mode finite-displacement calculation on the starter geometry produces the temperature dependence in Table 6.1.

T (K)	C_Q (MHz)	η	lines (MHz)
static	−5.173	0.043	0.11, 3.82, 3.94
0	−4.963	0.016	0.04, 3.70, 3.74
77	−4.927	0.014	0.03, 3.68, 3.71
150	−4.816	0.002	0.00, 3.61, 3.61
300	−4.534	0.033	0.08, 3.36, 3.44

Table 6.1: Finite-temperature ^{14}N EFG of NaNO_2 from a six-mode harmonic average on the starter geometry. The slope of the two strong lines is $d\nu/dT \approx -1.0$ to -1.1 kHz/K over 0–300 K.

The result reproduces the expected physics with no fitting: $|C_Q|$ and the lines decrease smoothly with temperature (the Bayer sign), the 0 K value already lies below the static one by about 4% (zero-point motion), and η shifts with temperature. The slope, ~ -1.0 kHz/K, has the correct sign and order of magnitude relative to the measured coefficients of -1.3 kHz/K (ν_-) and -2.2 kHz/K (ν_+).

Caveat. The numbers above are for the *unrelaxed* starter geometry: its static $\eta = 0.04$ disagrees with the experimental 0.38, the two strong lines come out near-degenerate (~ 3.7 MHz) rather than well split (3.60/4.64 MHz), and the phonon spectrum contains imaginary modes. The pipeline is therefore validated as physically sensible, but quantitative agreement requires relaxing the structure to an energy minimum first — run Stage 0 (§5.2) and pass the resulting `relaxed.abi` as the base input to remove the imaginary modes and correct η before interpreting the absolute numbers.

Runnable comparison. `examples/abinit/nano2_relaxation_study.sh` automates exactly this check: it runs the full temperature workflow on both the unrelaxed and the relaxed geometry and writes a side-by-side report (static EFG, the sweep, and $d\nu/dT$) scored against the measured NaNO_2 ^{14}N reference. See `examples/abinit/README_relaxation_study.md` for the prerequisites and the one-command invocation.

Relaxation outcome. Running this on the starter geometry confirms the pipeline behaves correctly and is qualitatively improved by relaxation, but also exposes its current limit. Relaxing the internal coordinates removes all imaginary modes (the spectrum becomes dynamically stable, lowest libration $\sim 44\text{cm}^{-1}$) and fixes the sign of every $d\nu/dT$, and it moves the asymmetry and the line splitting toward experiment — but only about a third of the way ($\eta : 0.04 \rightarrow 0.14$ against the measured 0.38; strong-line splitting $0.11 \rightarrow 0.37$ MHz against ~ 1.04 MHz). Because the cell was held fixed at the hand-entered starter (`optcell 0`, unconverged `ecut/pawecutdg/ngkpt`), the residual error sits in the *equilibrium* geometry, not the vibrational average. Quantitative agreement therefore needs a converged calculation on an experimental structure (e.g. the ICSD 82857 input under `structures/NaNO2/generated/`), and, for the full slope magnitude, anharmonic (AIMD/PIMD) averaging near the ferroelectric transition.

7 Validation Against the Measured Database

The cross-project `mr_integration` layer closes the loop against the `NQRDatabase`. `measured_temperature_coefficients` reads the stored $d\nu/dT$ for a compound and isotope; `compare_temperature_coefficients` matches a predicted set of (frequency, $d\nu/dT$) pairs — from either a `efg_temperature_sweep` (via `slopes_from_temperature_points`) or a Bayer fit — to the measured coefficients by frequency, so that a DFT temperature dependence can be checked against experiment. The same layer also validates the static (C_Q, η) against measured line positions and supplies the $C_Q \leftrightarrow \nu_Q$ conversion used by the simulator.

8 Public API Reference

8.1 `quadrupolar__dft.tensors / quadrupolar / abinit`

- `EFGTensor`, `average_tensors` — tensor conventions and frame-correct averaging.
- `coupling_constant_hz`, `nqr_frequencies_hz` — C_Q and zero-field transition frequencies.
- `AbinitEFGRecord`, `parse_abinit_efg`, `format_abinit_efg_block` — ABINIT EFG input/output.

8.2 `quadrupolar__dft.thermal`

- `thermal_quantum_factor`, `bose_occupation`, `mean_square_normal_coordinate`, `wavenumber_to_angular_frequency`

8.3 `quadrupolar__dft.vibrational`

- `VibrationalMode`, `efg_curvature_central_difference`, `thermally_averaged_efg`, `efg_temperature_sweep`, `ThermalEFGPoint`.
- `bayer_frequency`, `bayer_slope_hz_per_k`, `fit_bayer_single_mode`, `BayerFit`.

8.4 `quadrupolar__dft.finite__displacement / abinit__phonon`

- `Crystal`, `PhononMode`, `parse_abinit_structure`, `generate_displacement_jobs`, `abinit_input_with_positions`, `write_jobs`.
- `collect_efg_outputs`, `vibrational_modes_from_collected`, `vibrational_modes_from_efg`, `efg_tensor_from_record`.
- `phonon_dfpt_input`, `anaddb_input`, `parse_anaddb_modes`, `modes_from_arrays`.

8.5 `quadrupolar__dft.relax`

- `relax_input` (ionic-relaxation input from a static EFG input), `parse_relaxed_structure` (relaxed geometry from the ABINIT output footer), `relaxed_input` (relaxed static EFG input for Stages 1–2).

9 Limitations and Roadmap

Current limitations.

- The $C_Q \leftrightarrow \nu_Q$ scale and several helpers are calibrated for spin 1 and spin 3/2; higher half-integer spins are not yet handled by the simulator-facing conversion.
- The generated DFPT input is a template; PAW DFPT settings, tolerances and k -mesh must be checked for production accuracy.
- The anaddb eigenvector parser targets ABINIT 9.x text output; a `-modes` JSON escape hatch covers other layouts.
- Finite-difference curvatures use the printed EFG precision; large displacement amplitudes or extra output digits improve the second derivative.

Roadmap.

- Full cell relaxation (`optcell`) and a quasi-harmonic thermal-expansion path, building on the internal-coordinate relaxation stage (§5.2); plus AIMD/PIMD snapshot averaging for strongly anharmonic systems such as NaNO_2 near its ferroelectric transition.
- A `netCDF` / `phonopy` phonon reader to complement the `anaddb` text parser.